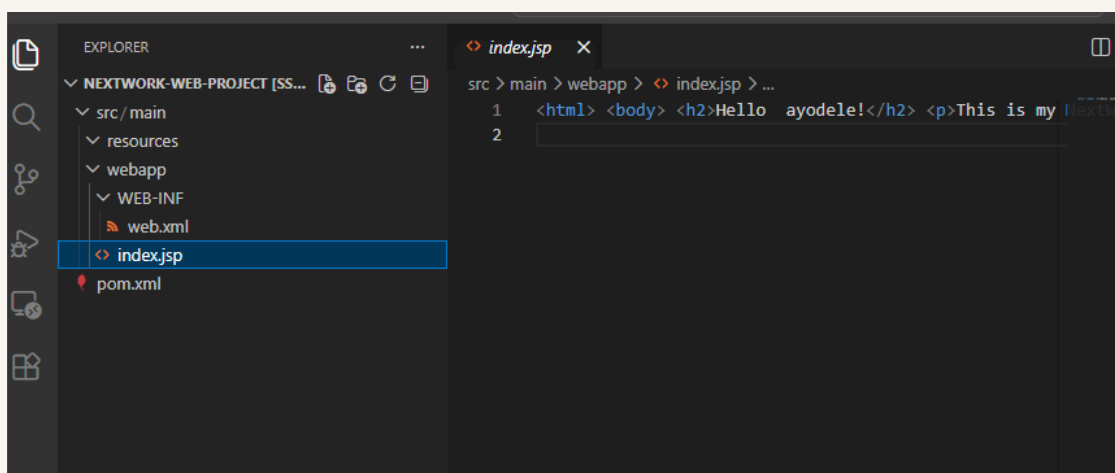




Set Up a Web App Using AWS and VS Code

GE

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Introducing Today's Project!

This project is Day One of the seven days DevOps challenge. In this project, we are going to set the foundation of CI/CD pipeline by creating a web app from scratch. We will need to launch an EC2 instance and connect to the instance through VS code to generate a web app inside.

This project is part one of a series of DevOps projects where I'm building a complete CI/CD pipeline. The first mission focused on editing index.jsp via the terminal, connecting to a remote server with SSH, and verifying changes in VS Code.

I did this project because it was the perfect opportunity to practice hands-on DevOps skills and reinforce concepts that go beyond theory.

Key tools and concepts

Services I used were AWS EC2 — to host and run the application remotely. SSH — for secure login and remote file editing using key pairs. VS Code — to verify changes and manage project files locally. Terminal Editors (nano/vim) — for direct editing of index.jsp without an IDE. Key concepts I learnt include...

Project reflection

This project took me approximately four hours. The most challenging part was connecting to my instance. It was most rewarding to complete this project.

One thing you probably didn't expect in this project was how file permissions and SSH key management could become such a critical hurdle. At first glance, connecting to an EC2 instance with a .pem key seems straightforward, but the "Permission denied" error revealed how sensitive cloud environments are to proper configuration.

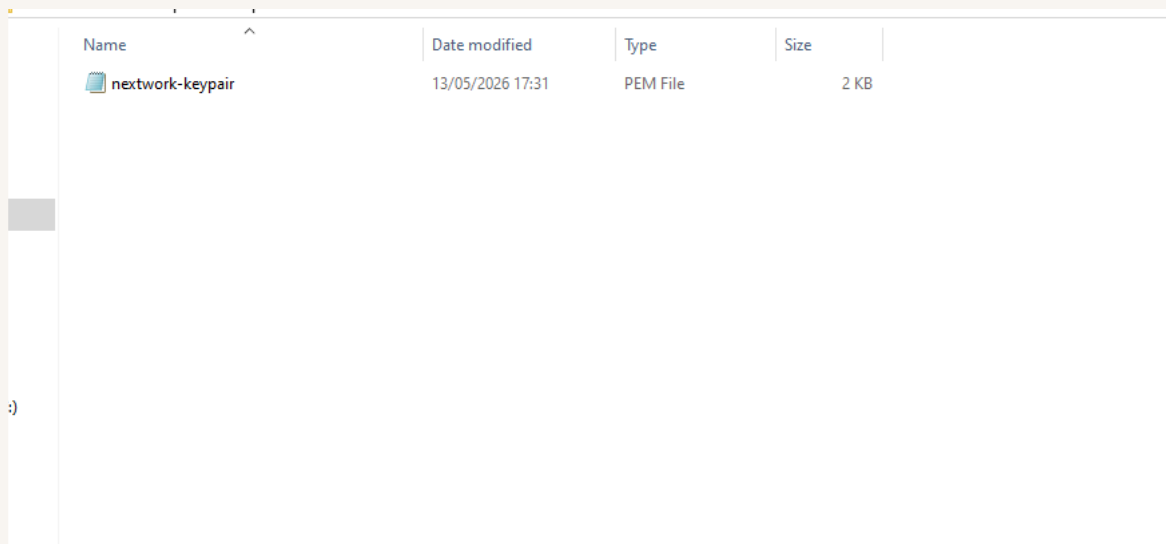


Launching an EC2 instance

What I did in this step

In this step, I will launch a new EC2 instance, set up a key pair, and configure network settings because I need a secure, accessible virtual server in AWS that is protected against unauthorized access while still being able to communicate safely with the internet and other resources.

I started this project by launching an EC2 instance because I needed a virtual server in AWS that provides scalable computing power, allowing me to run applications and workloads without relying on physical hardware. To secure access, I also set up a key pair because it ensures only authorized users with the private key can connect via SSH. Finally, I configured the network settings because defining security groups and VPC rules controls how the instance communicates with the internet and other resources, balancing accessibility with protection.



Name	Date modified	Type	Size
 nextwork-keypair	13/05/2026 17:31	PEM File	2 KB



I also enabled SSH

SSH (Secure Shell) is a secure network protocol that lets you remotely access and manage servers over the internet while keeping all communication encrypted. It's the standard way to connect to your EC2 instance safely.

Key pairs

A key pair is a set of two mathematically linked cryptographic keys — a public key and a private key — used in asymmetric encryption to secure communication, authenticate identities, and verify digital signatures. The public key is openly shared, while the private key must remain secret; together, they enable secure data exchange and trust in digital systems.

Downloaded key pair file

Once I set up a key pair in AWS, the platform automatically generates two linked keys — a public key and a private key. The public key is stored in AWS and associated with your resources (like EC2 instances), while the private key is immediately downloaded to your computer as a .pem file.



Set up VS Code

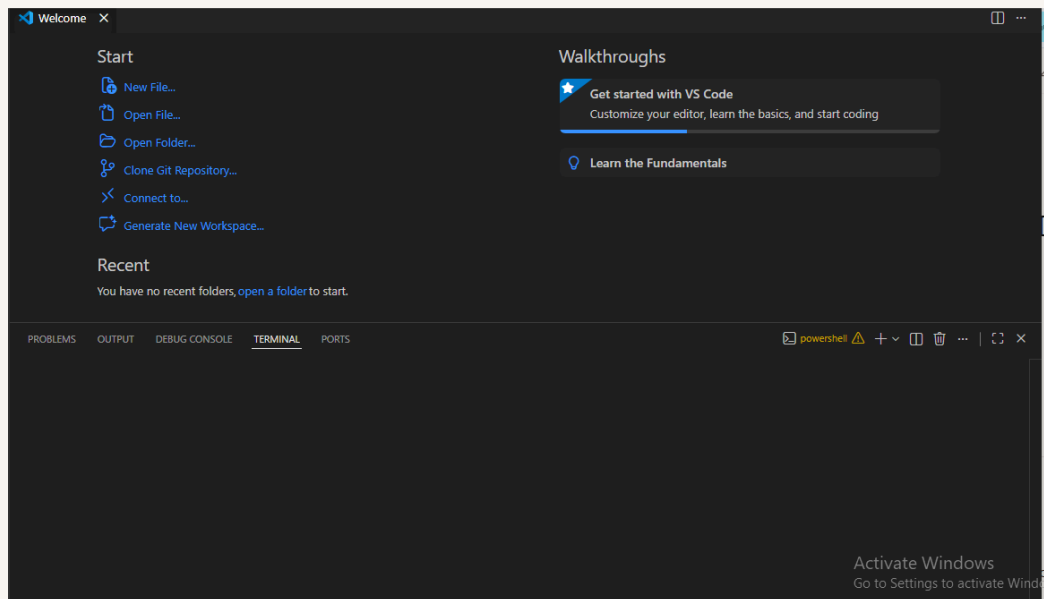
What I did in this step

In this step, I will download VS Code, set up a terminal in VS Code, and update my key pair's permission settings because I need a secure and convenient environment to communicate with my EC2 instance. VS Code provides an integrated terminal for running SSH commands, and updating the key pair permissions ensures that my private key is properly secured so I can log in to my instance without errors.

What is VS Code?

Visual Studio Code (VS Code) is a free, lightweight yet powerful source code editor developed by Microsoft, widely used for programming, debugging, and managing projects across many languages. It runs on Windows, macOS, and Linux, and is especially popular for web and cloud development.

I installed VS Code because it gives me a powerful and convenient environment to manage my project. With its integrated terminal, I can run SSH commands to connect to my EC2 instance directly inside the editor. It also allows me to edit and manage code, use extensions for AWS, and handle version control seamlessly, all in one place. This makes my workflow more secure, efficient, and organized.





My first terminal commands

A terminal is a tool that lets you type commands directly into your computer to interact with the operating system or remote servers. Think of it as a text-based interface where you can control your machine or connect to others without using graphical menus. The first command I ran for this project is 'cd ~\Desktop\DevOps'

Updating file permissions

I also updated my private key's permissions by running this code 'chmod 400 my-key.pem'

```
Directory: C:\Users\acer\Desktop\DevOps

Mode                LastWriteTime         Length Name
----                -
-a-----         13/05/2026      17:31           1678 nextwork-keypair.pem

PS C:\Users\acer\Desktop\DevOps> whoami
desktop-8l7p9ps\acer
PS C:\Users\acer\Desktop\DevOps> icacls "nextwork-keypair.pem" /reset
processed file: nextwork-keypair.pem
Successfully processed 1 files; Failed processing 0 files
PS C:\Users\acer\Desktop\DevOps> icacls "nextwork-keypair.pem" /grant:r "acer:R"
processed file: nextwork-keypair.pem
Successfully processed 1 files; Failed processing 0 files
PS C:\Users\acer\Desktop\DevOps> icacls "nextwork-keypair.pem" /inheritance:r
processed file: nextwork-keypair.pem
Successfully processed 1 files; Failed processing 0 files
PS C:\Users\acer\Desktop\DevOps> 
```



SSH connection to EC2 instance

What I did in this step

In this step, I will connect to my EC2 instance because I need secure remote access to the server in order to run commands, manage applications, and configure resources directly from my computer. By using SSH with my key pair, I ensure that only authorized access is allowed, keeping my cloud environment safe while giving me full control over the instance.

Connecting to EC2

To connect to my EC2 instance, I ran the command `ssh -i "C:\Users\acer\Desktop\DevOps\network-keypair.pem" ec2-user@ec2-34-204-73-93.compute-1.amazonaws.com`

This command required an IPv4 address

A server's IPv4 DNS usually refers to the Domain Name System address associated with the server's IPv4 address. It's the "phonebook of the internet," translating human-readable domain names (e.g., `example.com`) into machine-friendly IP addresses so your server or device can locate websites.



```
processed file: nextwork-keypair.pem
Successfully processed 1 files; Failed processing 0 files
PS C:\Users\acer\Desktop\DevOps> ssh -i C:\Users\acer\Desktop\DevOps\network-keypair.pem ec2-user@ec2-34-204-73-93.compute-1.amazonaws.com
Warning: Identity file C:\Users\acer\Desktop\DevOps\network-keypair.pem not accessible: No such file or directory.
The authenticity of host 'ec2-34-204-73-93.compute-1.amazonaws.com (34.204.73.93)' can't be established.
ED25519 key fingerprint is SHA256:M2q5e4upLR91P06YmmUVizni8IKs1K6or5iG9QRX/xo.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-34-204-73-93.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
ec2-user@ec2-34-204-73-93.compute-1.amazonaws.com: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
PS C:\Users\acer\Desktop\DevOps> ssh -i "C:\Users\acer\Desktop\DevOps\network-keypair.pem" ec2-user@ec2-34-204-73-93.compute-1.amazonaws.com

#_
#####
~\  #####
      \#####
      \###|
      \#/  https://aws.amazon.com/linux/amazon-linux-2023
      V#  '->
      /#  /
      /m  /
      /m  /

[ec2-user@ip-172-31-40-50 ~]$
```

Activate Windows
Go to Settings to activate Windows



Maven & Java

What I did in this step

In this step, I will install Apache Maven, install Amazon Corretto 8, and verify the installations because I need a complete Java development environment on my EC2 instance. Maven will manage dependencies and automate builds, while Corretto provides a stable version of Java to compile and run applications. Verifying ensures both tools are correctly installed and ready for use in my project.

Why I'm using Maven

Apache Maven is a powerful open-source build automation and dependency management tool, primarily used for Java projects. It simplifies compiling, testing, packaging, and deploying applications by using a standardized XML configuration file called the Project Object Model (POM).

Maven is required in this project because it provides a standardized, reliable way to manage builds and dependencies, which saves a lot of manual effort and reduces errors.

Why I'm using Java

Java is a high-level, object-oriented programming language created in 1995 by James Gosling at Sun Microsystems (now owned by Oracle). It is widely used because of its “write once, run anywhere” principle, meaning compiled Java code can run on any system with a Java Virtual Machine (JVM).



We're using Java in this project because it offers a combination of stability, portability, and a mature ecosystem that aligns well with the project's needs.



Create the Application

What I did in this step

In this step, I will generate the initial skeleton of a Java web application using Maven's archetype plugin, because it provides a ready-made project template with the correct structure and configuration files for a web app.

Creating the Java web app

I generated a Java web app using the command `mvn archetype:generate -DgroupId=com.nextwork.app -DartifactId=nextwork-web-project -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false`

Installing Remote - SSH

I installed Remote - SSH because it allows me to securely connect from my local development environment (like VS Code) to a remote server and work on it as if it were local.



SSH configuration details

Configuration details required to set up a remote connection include host Name: The IP address or domain name of the remote server (e.g., 192.168.1.10 or myserver.example.com). User Name: The account you'll log in with on the remote machine (e.g., ubuntu, root, or your personal username). Port: The SSH port (default is 22, unless your server uses a custom port). Authentication Method: Either a password or, more commonly, an SSH key pair (id_rsa private key stored locally, public key added to the server). Identity File: Path to your private key file if you're using key-based authentication (e.g., ~/.ssh/id_rsa). Remote Workspace Path: The directory on the server where your project files are located.

```
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp/1.0/maven-archetype-webapp-1.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp/1.0/maven-archetype-webapp-1.0.jar (3.9 kB at 69 kB/s)
[INFO] -----
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] -----
[INFO] Parameter: basedir, Value: /home/ec2-user
[INFO] Parameter: package, Value: com.nextwork.app
[INFO] Parameter: groupId, Value: com.nextwork.app
[INFO] Parameter: artifactId, Value: nextwork-web-project
[INFO] Parameter: packageName, Value: com.nextwork.app
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/ec2-user/nextwork-web-project
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 18.921 s
[INFO] Finished at: 2026-05-13T18:03:58Z
[INFO] Final Memory: 17M/61M
[INFO] -----
[ec2-user@ip-172-31-40-50 ~]$
```

Activate Windows
Go to Settings to activate Windows

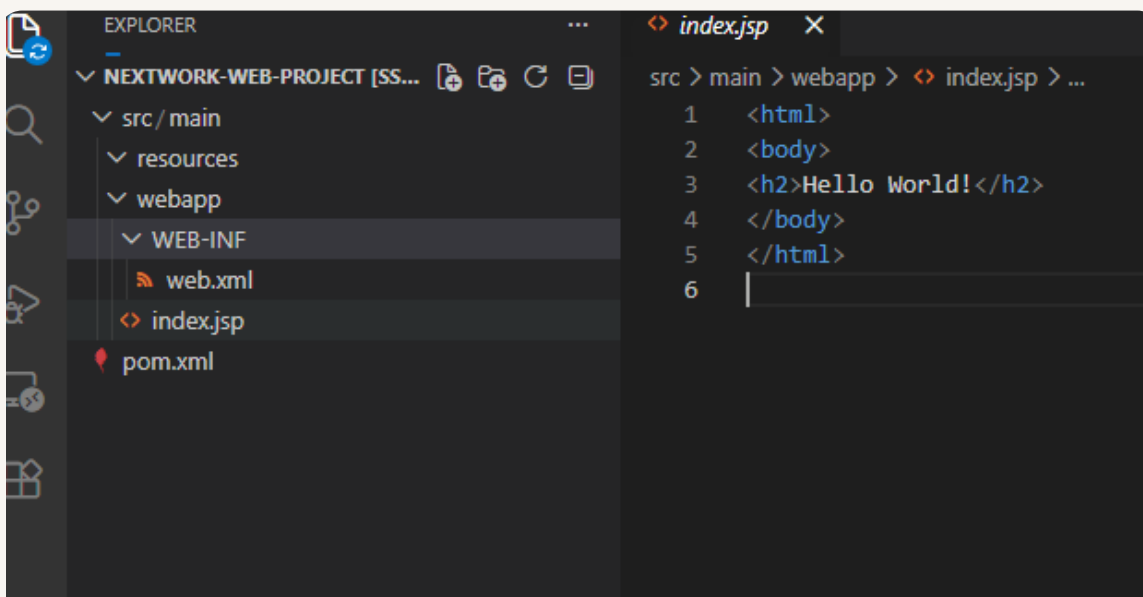


Create the Application

Exploring the project structure

Using VS Code's file explorer, I could see the my-key.pem file sitting in my workspace directory, alongside other project items such as source code files (.js, .py, .ts), configuration files (package.json, requirements.txt), and hidden files like .gitignore or .env. I could also see folders representing the project structure — for example src/, assets/, or tests/ — all neatly organized in the tree view on the left sidebar.

Two of the project folders created by Maven are src and webapp, which are the src folder usually contains your application's source code (logic, components, styles, utilities), while the webapp folder typically holds the web application files that are deployed to a server, such as JSPs, HTML, CSS, and JavaScript. Together, they separate development code from deployable assets.





Using Remote - SSH

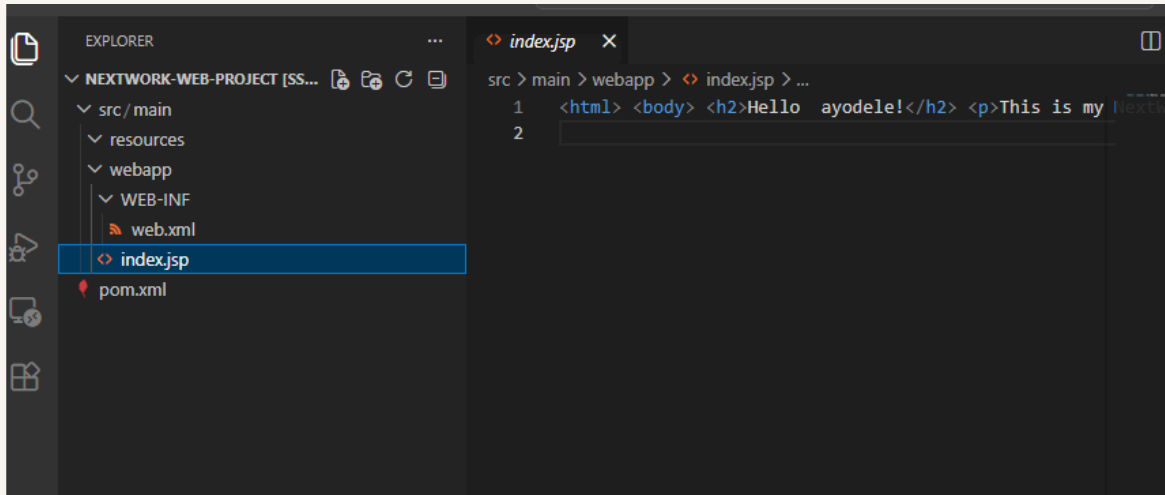
What I did in this step

In this step, I will install an extension in VS Code, use the extension to set up a connection between VS Code and my EC2 instance, and explore and edit my Java web app's files because I want a seamless workflow where I can manage my cloud server directly from my editor. The extension makes it easier to establish a secure connection, and once connected, I can browse, open, and modify my project files inside VS Code without needing to manually transfer them. This setup improves efficiency, keeps everything in one place, and allows me to develop and maintain my Java web app more effectively.

Updating the web app

An index.jsp file is a JavaServer Pages (JSP) file that serves as the default landing page for a web application directory, acting similarly to a default index.html file but allowing for dynamic server-side Java code.

I edited index.jsp by pasting the codes on text editor and include my name.





Using nano

Additional improvements

In this secret mission, I will edit `index.jsp` using the terminal instead of an IDE, because I want to demonstrate mastery of command-line tools and showcase adaptability in environments where graphical interfaces aren't available.

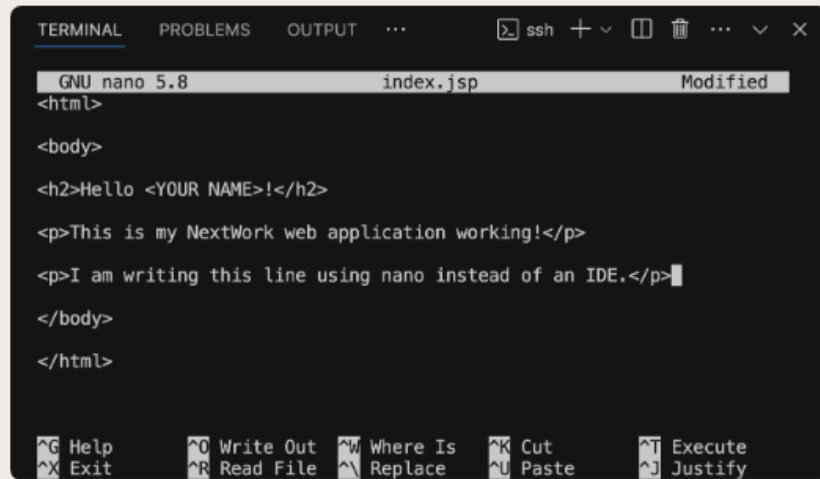
Terminal vs IDE

An alternative to using IDEs is working directly from the terminal, which allows developers to edit files, run commands, and manage projects without relying on graphical tools. This approach is lightweight, efficient, and ideal for remote or server-based environments.

Editing code in the terminal is lightweight, fast, and ideal for remote or resource-limited environments, while IDEs provide a full suite of tools (debugging, compiling, version control) that streamline complex development. The choice depends on whether you value speed and simplicity or integrated features and productivity.

Verifying my work

To verify my editing work in the terminal, I saved the changes in nano (or vim) and then reopened the project in VS Code. Because VS Code continuously monitors the project folder, any file changes made externally (like through the terminal) are immediately reflected in the editor. This made it possible to see my updates to `index.jsp` right away without needing to restart or reload the IDE.



The screenshot shows a terminal window with a dark background. At the top, there are tabs for 'TERMINAL', 'PROBLEMS', and 'OUTPUT'. The terminal title bar indicates an 'ssh' session. The nano editor's status bar at the top shows 'GNU nano 5.8', the filename 'index.jsp', and the state 'Modified'. The editor content consists of an HTML document structure with a body containing two paragraphs. The first paragraph says 'Hello <YOUR NAME>!' and the second says 'This is my NextWork web application working!'. The second paragraph is followed by a line indicating the use of nano instead of an IDE. The bottom of the window features a row of keyboard shortcuts for various nano commands like Help, Exit, Write Out, Read File, etc.

```
TERMINAL  PROBLEMS  OUTPUT  ...  ssh + - [] [] ... - x
GNU nano 5.8  index.jsp  Modified
<html>
<body>
<h2>Hello <YOUR NAME>!</h2>
<p>This is my NextWork web application working!</p>
<p>I am writing this line using nano instead of an IDE.</p>
</body>
</html>

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```



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